Actual quadratic-twist counts and a literal isomorphism counterexample

Result. Lean proves the actual count formula for every finite odd field and nonsingular cubic model $y^2=x^3+a\cdot x^2+b\cdot x+c$: nonsquare scaling $(d\cdot a,d^2\cdot b,d^3\cdot c)$ gives $N+N'=2(q+1)$, actual trace $t'=-t$ and equal source gcd(trace,q) predicates. It also supplies an explicit counterexample to the literal non-isomorphism assertion in Galbraith Exercise9.5.3. [pcc:unverified]

Valid count theorem	Every finite odd field; nonsquare d, $\Delta\neq 0$ Actual $N+N'=2(q+1)$, $t'=-t$; source predicates agree
Literal Ex9.5.3	F7, j=1728, nonsquare d=3; both #E=8 Isomorphism (4X:Y:Z), inverse (2X:Y:Z)
Delete hypotheses	F3, d=1: actual sum 10 ≠ 8 F5 singular cusp, nonsquare d=2: sum 10 ≠ 12

Actual semantics. Actual points are equated with literal equation pairs plus the identity, then with square-root fibers. Zero has one root, a nonzero square two, a nonsquare none. Official finite-field square criteria prove complementary fibers for v and d^3v have total two. Reindex x by $d\cdot x$, add the two identities and derive actual counts/traces. Discriminants scale by d^6. No non-isomorphism, Frobenius trace or point-count premise is assumed. This proves the source short-model count exercise and the explicit cubic-with-x^2 extension. [pcc:unverified]

Literal primary counterexample. Over F7, $y^2=x^3+x$ is nonsingular with j=1728. The nonsquare d=3 gives $y^2=x^3+2x$, yet $(X:Y:Z)\mapsto(4X:Y:Z)$ has inverse (2X:Y:Z). Lean proves the actual admissible variable change, actual additive point-group equivalence, both counts eight, and the homogeneous cubic/inverse identities over every commutative ring of characteristic seven. This is a trivial nonsquare scaling in the special-j case. The separately qualified generic-j Lemma9.5.7 retains its explicit exceptions. [pcc:unverified]

Hypothesis controls. Deleting nonsquare scaling fails on the credited F3 curve with five actual points: d=1 gives sum10 rather than8. Deleting nonsingularity fails on the credited F5 cusp: nonsquare d=2 leaves both actual nonsingular point groups of count5, giving10 rather than12. Both are direct universal negations. F7 addition compatibility is a closed kernel check on actual point types through a proved equation-point equivalence. [pcc:unverified]

Primary review. Galbraith author version2 chapter9, SHA f9a3272a..., selected Theorem9.3.4/Definition9.5.1/Example9.5.2/Exercises9.5.3–9.5.4; printedp171/PDF7 fully rendered and inspected. Author landing/errata checked, including adjacent9.5.8 correction; no novelty or previously-unreported-error claim. FST selected trace/7.3 background; Milne IV7 restriction char≠2,3 retained. KB/Conductor unavailable; bounded primary/official/prior fallback recorded. No full-book read or corpus promotion. [pcc:unverified]

Validation and scope. Whole gate 42 declarations, zero failures, zero trusted native exceptions. All 64 build/provenance files retained losslessly in 25 objects; five gate files, complete statement and recursive source/config pins. Lean4.35.0/Mathlib56a00585..., standard foundations only. Exact local PCC replay differs from signed standing. General characteristic-three trace-to-geometric criterion is the next component; general Hasse/Endrank/embedding/full-torsion-field and remaining original fourteen-item gaps stay open. No holes/custom axioms/native evaluator/raised limits/new global flags/waivers/native experiment/paid compute/agent upload. [pcc:unverified]